

MISSOURI
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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 7.1

Diagonalization of
Symmetric Matrices

Symmetric Matrices

A matrix A is symmetric if and only if $A^T = A$.

Example 1

Show that if A is symmetric then A^2 is symmetric.

Example 2

Diagonalize the matrix

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 5 & 1 \\ 5 & 1 & 1 \end{bmatrix}$$

given that A has eigenvalues -4 , 4 , and 7 .

Theorem

If A is symmetric, then any two vectors from different eigenspaces are orthogonal.

Example 3

Prove the preceding theorem.

Orthogonal Diagonalizability

An $n \times n$ matrix A is orthogonally diagonalizable if there exists an orthogonal matrix P and a diagonal matrix D such that

$$A = PDP^{-1} = PDP^T$$

An $n \times n$ matrix A is orthogonally diagonalizable if and only if A is a symmetric matrix.

Example 4

Diagonalize the matrix

$$A = \begin{bmatrix} 5 & 8 & -4 \\ 8 & 5 & -4 \\ -4 & -4 & -1 \end{bmatrix}$$

given that A has eigenvalues -3 and 15 .

The Spectrum of a Matrix

The set of the eigenvalues of a matrix A is sometimes called the spectrum of A .

Theorems dealing with the eigenvalues of a matrix are sometimes called spectral theorems.

The Spectral Theorem for Symmetric Matrices

An $n \times n$ symmetric matrix A has the following properties:

- A has n real eigenvalues, counting multiplicities.
- The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.

The Spectral Theorem for Symmetric Matrices

An $n \times n$ symmetric matrix A has the following properties:

- The eigenspaces of A are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.
- A is orthogonally diagonalizable.

Spectral Decompositions

Let A be an $n \times n$ symmetric matrix.

Then, $A = PDP^{-1} = PDP^T$ where the columns of P are orthonormal eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ corresponding to eigenvalues $\lambda_1, \dots, \lambda_n$.

A spectral decomposition of A is

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \dots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T$$

Example 5

Construct a spectral decomposition of

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Comments on Spectral Decompositions

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \dots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T$$

Every term in the spectral decomposition is an $n \times n$ matrix of rank 1.

Each matrix $\mathbf{u}_j \mathbf{u}_j^T$ is a projection matrix – in other words, for each $\mathbf{x}$ in $\mathbb{R}^n$, the vector $(\mathbf{u}_j \mathbf{u}_j^T) \mathbf{x}$ is the projection of $\mathbf{x}$ onto $\mathbf{u}_j$.
